



Retail Revenue Optimization & Customer Lifetime Value (CLV) Analysis

i A proactive, end-to-end analytical pipeline transforming 12,500+ rows of raw, uncleaned transactional retail data into high-value strategic business recommendations.

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Executive Summary & Project Background

- **Context:** This proactive case-study simulates an end-to-end performance review for a multi-category retail store.
- **The Mission:** To transform an uncurated, messy retail dataset into strategic insights using Excel exclusively. This was achieved by applying algebraic formulas to fix missing data, performing manual cross-checks to guarantee accuracy, establishing core financial KPIs, segmenting customers via Customer Lifetime Value (CLV) and deploying a fully interactive Excel dashboard for executive decision-making.
- **Keys Results Overview:** Transformed a highly corrupted dataset into a validated source of truth, establishing that **64% of the variance in Customer Lifetime Value (CLV) aligns with Purchase Frequency ($r = 0.80$)** and deployed an interactive Excel dashboard for real-time stakeholder decision-making.

Dataset Overview

- **Source:** [Ahmed Mohamed \(Kaggle\) — Retail Store Sales \(Dirty for Data Cleaning\)](#).
- **Scale:** 12,500+ transactional records.
- **Characteristics:** Explicitly engineered with systemic errors, missing values (Price , Quantity), and categorical inconsistencies to simulate a raw, uncured corporate database.

Data Acquisition & Sourcing

- **Sourced and extracted** a raw retail transaction dataset from Kaggle to conduct performance and customer behavior analysis.

Data Cleaning & Standardization

- **Standardized Data Types:** Converted numerical columns (Price Per Unit , Quantity , Total Spent) from text format to numerical values by replacing localized delimiter punctuation (dots to commas), enabling accurate Excel mathematical operations.
- **Resolved Primary Key Integrity:** Audited the dataset using the Transaction ID primary key to confirm zero duplicate records, ensuring data uniqueness.

Advanced Missing Value Imputation (Logical & Statistical)

- **Deterministic Financial Reconstruction:** Imputed missing values in Price Per Unit and Total Spent by mathematically reconstructing the rows using the structural relationship formula: $\text{Total Spent} = \text{Quantity} \times \text{Price Per Unit}$.
- **Contextual Product Deduction:** Deduced missing Item names by cross-referencing available Price Per Unit data against known product category code combinations.
- **Statistical Risk Mitigation for Quantity:** Analyzed central tendency metrics (Mode vs. Median) per product category to handle missing Quantity values. Rejected the statistical Mode (10) as a retail outlier, electing to impute using the **Category Median** (6). This statistical choice protected the central tendency and prevented artificial revenue inflation in subsequent financial reporting.
- **Categorical Handling:** Designated missing records in the Discount Applied boolean column as "Unknown" due to insufficient contextual data, preserving data auditability without introducing bias.

Data Quality Assurance & Validity Checks

- **Formula Validation:** Conducted a comprehensive validity audit across all existing records by verifying that pre-existing Total Spent values mathematically matched the calculated product of Quantity and Price Per Unit.

Feature Engineering & Aggregation

- **Time-Series Derivation:** Extracted `Day of the Week` and `Month of the Transaction` from raw timestamps to enable temporal trend analysis.

KPIs & Customer Lifetime Value (CLV) Analysis

- This section translates the cleaned transactional data into high-level financial KPIs and customer-level valuations within a unified workspace.

High-Level Aggregate KPIs

Using aggregated Excel formulas on the cleaned master dataset, 6 baseline performance metrics were established to track store health.

- **Total Revenue:** Measures total client spend, representing the company's gross earnings.
- **Total Units Sold:** Tracks the total volume of inventory moved across all categories.
- **Total Transactions:** Measures the total number of processed orders.
- **Average Order Value (AOV):** Quantifies the standard customer basket value (calculated at **€130.08**).
- **Average Items per Basket:** Measures the typical volume of products per transaction (**6 items**).
- **Average Price per Item:** Tracks the mean price across all products sold to monitor pricing benchmarks.

Customer-Level Equity Modeling

*To find long-term strategic insights, raw transactional data was aggregated by unique **Customer IDs (T_01 to T_17)** into a customer-level summary table to evaluate individual purchasing behaviors:*

- **Revenue Contribution (Summe von Total Spent):** Evaluated the total capital spent by each individual customer.
- **Transaction Velocity (Anzahl von Transaction ID):** Monitored the exact number of orders placed by each specific customer.
- **Behavioral Synthesis (AOV & Purchase Frequency):** Tracked the personal average ticket size and returning visit frequency per customer.
- **Customer Lifecycle Value (CLV):** Combined these individual metrics to calculate the final long-term financial valuation of each unique shopper.

Dashboard Layout & Co-Location

- **Visual Organization:** For clean workbook design, the high-level macro KPIs and the customer-level CLV table were co-located on the exact same Excel tab.

- **User Benefits:** While these two analytical blocks are mathematically independent and not connected, placing them on a single tab provides a unified workspace. This layout allows a business user to view overall store health and individual customer histories side-by-side without needing to click through multiple spreadsheet tabs.

Descriptive & Diagnostic Statistics

- This section translates the cleaned transactional data into high-level descriptive benchmarks, categorical performance metrics, and diagnostic behavioral correlations within a unified workspace.

Descriptive Statistics Findings

What happened: Establishing the baseline and behavior of the dataset.

- **The Average Basket:** Calculated a baseline transaction mean of **5.55 items** (Median: 6) with a mean expenditure of **130.08 €** per order.
- **Data Distributions:** Proved that unit pricing and item quantity follow a symmetrical distribution, while Total Spent and Customer Lifecycle Value (CLV) display a slight positive skew.

Diagnostic Statistics Findings

Why it happened: Uncovering the underlying relationships, revenue drivers, and category performance.

- **The Customer Lifetime Value (CLV) Engine:** Executed a correlation analysis proving a strong positive relationship ($r = 0.80$) between Purchase Frequency and Customer Lifetime Value (CLV), mathematically demonstrating that frequency accounts for 64% of total CLV variance ($(r^2 = 0.64)$).
- **Volume Anchors:** Isolated category performance to find that **Electric Household Essentials** and **Furniture** drive baseline transactional volume, sharing the peak volume at **1,591 units each**, closely followed by **Food** at 1,588 units.
- **Revenue Drivers:** Identified **Butchers** as the primary financial engine of the business, generating the highest total revenue segment at **216,480.50 €**.
- **Payment Behavior:** Segmented payment ecosystems to discover that Cash transactions yield the highest average basket value (**131.50 €**), marginally outperforming Credit Cards (129.54 €) and Digital Wallets (129.12 €).

Business Impact of the Analysis

- **Frequency Scales Value:** Proved mathematically that driving customer repeat-purchase frequency ($r = 0.80$) scales long-term account value far more efficiently than chasing isolated, large-scale single orders.

- **High-Yield Segments:** Identified Cash-paying accounts as premium transactional spenders (131.50 € mean spend), demonstrating that cash optimization remains a critical pillar alongside digital avenues.
- **Targeted Volume Strategies:** Isolated *Electric Household Essentials* and *Furniture* as the highest-volume customer entry points (1,591 units each), providing specific categories for targeted cross-selling campaigns to boost total revenue.

Business Intelligence (BI) Interface

- This unified workspace delivers an interactive view of the cleaned dataset—combining core business benchmarks, category sales performance, and diagnostic behavioral metrics.

System Design & Stakeholder Impact

- **Technical Implementation:** Built a fully interactive dashboard utilizing **Excel Slicers, Timelines, dynamic Pivot Charts, and hidden reference tables** to prevent visual clutter.
- **Interactive Capabilities:** Allowed users to filter macro-KPIs on-the-fly by **region, product category, and payment methods**.
- **Business Impact:** Empowered non-technical business stakeholders to **independently explore data trends and safely extract insights** without altering the underlying master dataset.

Key Performance Indicators (KPIs) Dictionary

- **Total Revenue (1.635.094,50 €):** Measures gross financial scale and top-line health.
- **Total Transactions (12,575):** Tracks total volume of orders processed.
- **Average Order Value (130,03 €):** Reflects the financial depth of individual checkouts.
- **Average Items Per Basket (6):** Tracks shopping basket density.

Interactive Slicer & Interface Elements

- **Left Slicer Panel:** Filters the entire dashboard by Day of the Week, Category, Payment Method, and Location (In-store vs. Online).
- **Top Timeline Slicer:** Controls the **Monthly Revenue Trend Line Chart** to track financial performance and seasonality across months 1–12.
- **Donut Chart:** Visualizes the categorical share of payment methods.
- **Horizontal Bar Chart:** Ranks product categories by revenue from highest (**Fresh Goods**) to lowest.

Advanced Statistics & Diagnostic Modeling

- **Customer Loyalty Correlation: 0,80**

- **Business Diagnostic:** A correlation coefficient of 0.80 indicates a strong positive relationship, where **64% of the variance** in Customer Lifetime Value (CLV) aligns with Purchase Frequency.

Strategic Deliverables (Inferences)

- **Pricing Strategy:** Recommended a **free-shipping threshold at €150** to systematically push the current €130.08 AOV higher.
- **Operational Optimization:** Advised **optimizing physical checkout cash-handling workflows** to support the 31.63% cash transactional volume.
- **Loyalty Focus:** Target customer retention programs specifically around increasing return visit frequency since it heavily dictates lifetime value.

Documentation Note: *This documentation was authored using a collaborative approach. The underlying data analysis, core insights, and analytical findings were entirely gathered and developed by the analyst. Generative AI was utilized strictly as an editorial tool to structure, polish, and finalize these personal notes into professional technical documentation.*